Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 5

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

Q1. Count Occurrences in a Tuple

Write a program to count the occurrences of a given element in a tuple.

Q2. Update a Tuple Element

Write a Python program to create a tuple containing the marks of five students. Accept a student position and a new mark from the user. Since tuples are immutable, convert the tuple into a list, update the mark at the specified position, and convert it back into a tuple. Display the original tuple and the updated tuple.

Q3. Find Common Elements in Two Sets

Write a program to find the common elements between two sets.

Q4. Unique Common Elements Using Tuple and Sets

Write a Python program to create two tuples containing the names of students who participated in Event 1 and Event 2. Some students may appear more than once in a tuple. Convert both tuples into sets and display:

1. The unique students who participated in both events.
2. The students who participated only in Event 1.
3. The total number of unique students who participated in either event.

Q5. Student Marks Analysis Using Tuple and Set

Write a Python program to create a tuple containing the names of five students and another tuple containing their corresponding marks. Create a set containing the marks of students who scored above 75. Display the set of qualifying marks and the number of students who scored above 75.

Q6. Student Grade Management Using Tuple and Set

Write a Python program to **accept the names and grades of five students from the user** and store them in two tuples.

1. Create a set containing the **unique grades** obtained by the students.
2. Count the number of students who received grade `'A'`.
3. If no student received grade `'A'`, display an appropriate message.
4. Display the student names, grades, unique grades, and the appropriate result for grade `'A'`.

Q7. Set Relationship Analysis

Write a program to create two sets of integers and determine whether the sets are equal, disjoint, or one is a subset of the other.

Q8. Employee Working Hours Analysis using Tuples

Write a Python program to accept the working hours of 7 employees from the user and store them in a tuple. Display the working hours of employees who worked more than 8 hours and count the number of employees who worked more than 8 hours.

Constraint: Do not use the map() function.

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